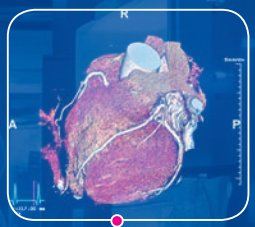
HEART DISEASE

Predicted using data on age, sex, blood pressure, cholesterol, calcium score, along with family history and lifestyle risks such as smoking, obesity



RARE DIABETES

About 2-5% patients of type-2 diabetes in India have MODY (Maturity-onset diabetes of the young), diagnosed only through genetic tests

THE AI ADVANTAGE

- **Early diagnosis:** Any prediction of a disease risk prompts increased monitoring/ screening, leading to an early diagnosis if the condition develops eventually
- **Better treatment outcomes:** An early diagnosis in turn leads to improved chances of treatment success
- **Less reliance on medication:** Doctors usually recommend lifestyle modifications, and not medicines, under predictive health programmes
- **Avoids surgery:** Timely lifestyle changes can prevent severe complications that usually result in surgery or heavy medication

WHAT ARE THE CHALLENGES?

- **False negatives:** A negative result does not guarantee that the person won't get the disease in the future
- **Limited data set:** For AI machines to make accurate predictions, a varied and large amount of data set is required to compare against the patient's results
- **Quality of diagnostics:** The analytics ultimately depends on blood markers, body scans and genetic markers. If these are not of high quality, the prediction will not be accurate

Powering this 'predictive healthcare' revolution is Artificial Intelligence, which has already permeated many aspects of human life and has come to healthcare too. AI, along with the Internet of Things, machine learning and other cutting-edge tech, has now spawned what is called the IoMT, or the Internet of Medical Things. AI produces algorithms out of vast amounts of data that help in precise and speedier diagnosis and treatment. The technology is already in widespread use in the West, where the prohibitive cost of treatment makes disease prediction a desirable ideal. It is now seeping into India too, as doctors and medical establishments begin to deploy the power of AI tools to predict and nip disease in the bud. At the same time, sundry players are engaged in making AI diagnostic tools that are more affordable and also scalable.

"It is essential that India is not left behind in the new predictive diagnostic technology," says NITI Aayog member Dr V.K. Paul. "By tapping into our domestic manufacturing potential, we can make new testing technology available to Indians at affordable rates. 'Make in India' will make diagnostics using AI and genetics more accessible." This will be especially useful in a country with a vast population like ours, where the allopathic doctor-to-patient ratio of 1:1,456 (per the Economic Survey 2019-20) is well below the World Health Organization-recommended figure of 1:1,000. Other statistics are similarly disheartening: access to quality healthcare remains extremely uneven as

"Data-backed analytics helps convince people to take care of their health... if you can **motivate them to take the necessary precautions,** it will reduce the incidence of disease and hospitalisations"

-DR NARESH TREHAN
Chairman, Medanta

